

Flow Descriptions and Flow Lines Homework

Student Version

Fluid Dynamics

Dr. Huan Phan

Robotics and Mechatronics Engineering, Randolph College

Use SI units unless otherwise stated. Show the differential equations and apply the stated initial or point conditions. Clearly distinguish between a streamline, a pathline, and a streakline.

Homework 1: Steady Two-Dimensional Flow Field

Problem

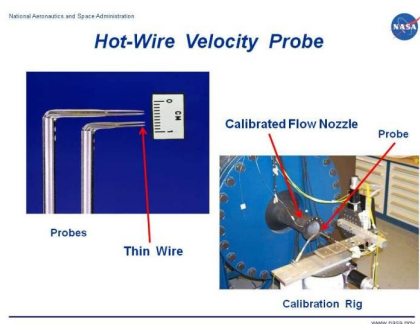
A steady two-dimensional velocity field is

$$u = 2x, \quad v = -2y,$$

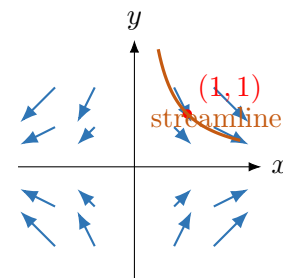
where velocity is in m/s when x and y are in meters. Determine:

1. the streamline passing through $(x, y) = (1, 1)$;
2. the pathline of a particle located at $(1, 1)$ at $t = 0$;
3. the particle position at $t = 0.50$ s;
4. whether the streamline and pathline have the same geometric curve.

Real measurement context



Velocity-field diagram



Steady field: $u = 2x$, $v = -2y$; particle begins at $(1, 1)$ when $t = 0$.

Hint

For a streamline, use

$$\frac{dy}{dx} = \frac{v}{u} = -\frac{y}{x}.$$

For the pathline, solve

$$\frac{dx}{dt} = 2x, \quad \frac{dy}{dt} = -2y$$

with $x(0) = 1$ and $y(0) = 1$.

Solution

The streamline equation is

$$\frac{dy}{dx} = -\frac{y}{x}.$$

Separate variables:

$$\frac{dy}{y} = -\frac{dx}{x}.$$

Integrating,

$$\ln y = -\ln x + C,$$

so

$$xy = C_1.$$

Because the streamline passes through $(1, 1)$, $C_1 = 1$. Therefore,

$$\boxed{y = \frac{1}{x}}.$$

For the particle pathline,

$$\frac{dx}{dt} = 2x \implies x = C_x e^{2t}.$$

Using $x(0) = 1$,

$$x = e^{2t}.$$

Similarly,

$$\frac{dy}{dt} = -2y \implies y = C_y e^{-2t}.$$

Using $y(0) = 1$,

$$y = e^{-2t}.$$

At $t = 0.50$ s,

$$x = e^1 = 2.718 \text{ m}, \quad y = e^{-1} = 0.368 \text{ m}.$$

Eliminating time gives

$$xy = e^{2t} e^{-2t} = 1,$$

which is the same geometric curve as the streamline because the flow is steady.

$$y = 1/x, \quad x(t) = e^{2t}, \quad y(t) = e^{-2t}, \quad (x, y)_{0.5s} = (2.718, 0.368) \text{ m}$$

Homework 2: Unsteady Uniform Flow

Problem

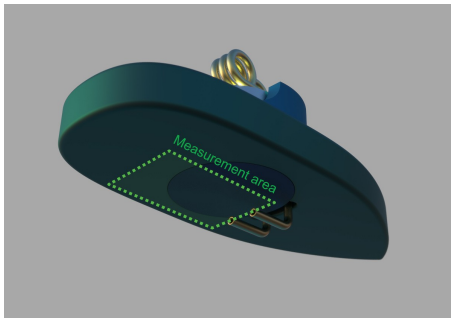
An unsteady two-dimensional flow has velocity components

$$u = 2.0 \text{ m/s}, \quad v = t \text{ m/s},$$

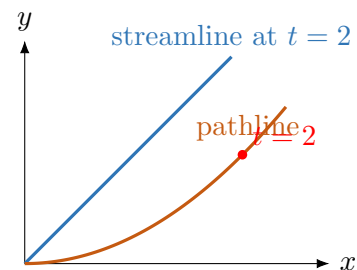
where t is measured in seconds. A particle is at the origin at $t = 0$. Determine:

1. the streamline through the origin at $t_* = 2.0$ s;
2. the particle pathline in parametric form;
3. the geometric equation of the pathline;
4. the particle position at $t = 2.0$ s.

Real measurement context



Flow-line diagram



Unsteady uniform field: $u = 2$, $v = t$; particle starts at $(0, 0)$ at $t = 0$.

Hint

Freeze time when calculating the streamline:

$$\left. \frac{dy}{dx} \right|_{t=t_*} = \frac{v(t_*)}{u}.$$

For the pathline, integrate

$$\frac{dx}{dt} = 2, \quad \frac{dy}{dt} = t.$$

Then eliminate t .

Solution

At $t_* = 2$ s,

$$\frac{dy}{dx} = \frac{2}{2} = 1.$$

The streamline through the origin is

$$\boxed{y = x}.$$

For the particle pathline,

$$\frac{dx}{dt} = 2 \quad \Longrightarrow \quad x = 2t + C_x.$$

Since $x(0) = 0$,

$$x = 2t.$$

Also,

$$\frac{dy}{dt} = t \quad \Longrightarrow \quad y = \frac{t^2}{2} + C_y.$$

Since $y(0) = 0$,

$$y = \frac{t^2}{2}.$$

Eliminate $t = x/2$:

$$y = \frac{1}{2} \left(\frac{x}{2} \right)^2 = \frac{x^2}{8}.$$

At $t = 2$ s,

$$x = 4.0 \text{ m}, \quad y = 2.0 \text{ m}.$$

The streamline and pathline differ because the flow is unsteady.

$$\boxed{\text{streamline: } y = x, \quad \text{pathline: } x = 2t, \quad y = t^2/2, \quad y = x^2/8, \quad (x, y)_{2s} = (4, 2) \text{ m}}$$

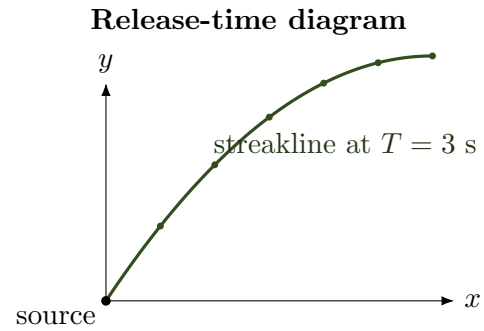
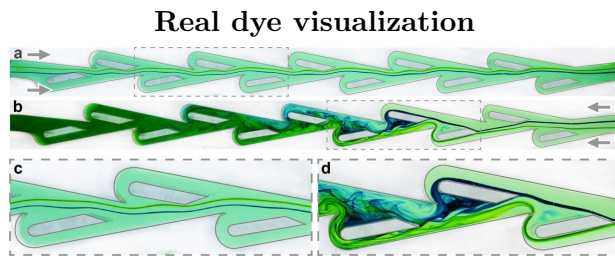
Homework 3: Dye Streakline from a Fixed Source

Problem

For the unsteady velocity field

$$u = 2.0 \text{ m/s}, \quad v = t \text{ m/s},$$

a dye source continuously releases particles from the origin beginning at $t = 0$. Determine the streakline observed at $T = 3.0$ s. Express the result as $y = y(x)$ and state the valid range of x .



Continuous source at the origin; $u = 2$, $v = t$; observation time $T = 3.0$ s.

Hint

Let τ be the release time. A particle released at $t = \tau$ satisfies

$$x = 2(T - \tau), \quad y = \int_{\tau}^T t \, dt = \frac{1}{2}(T^2 - \tau^2).$$

Eliminate τ using $\tau = T - x/2$. The earliest particle was released at $\tau = 0$.

Homework 4: Material Acceleration from an Eulerian Field

Problem

A steady velocity field is

$$u = 3x, \quad v = -2y.$$

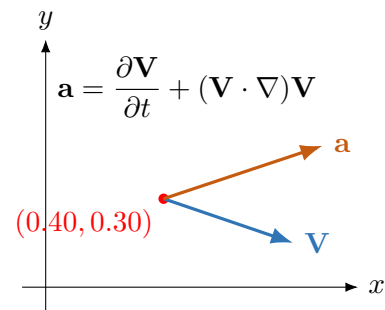
At the point $(x, y) = (0.40, 0.30)$ m, determine:

1. the local acceleration;
2. the convective acceleration components;
3. the total material acceleration vector.

Real flow-field measurement



Acceleration diagram



Steady field: $u = 3x$, $v = -2y$; evaluate at $(0.40, 0.30)$ m.

Hint

Because the flow is steady,

$$\frac{\partial u}{\partial t} = \frac{\partial v}{\partial t} = 0.$$

Use

$$a_x = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y}, \quad a_y = u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y}.$$

Homework 5: Pathline in a Rotating Uniform Flow

Problem

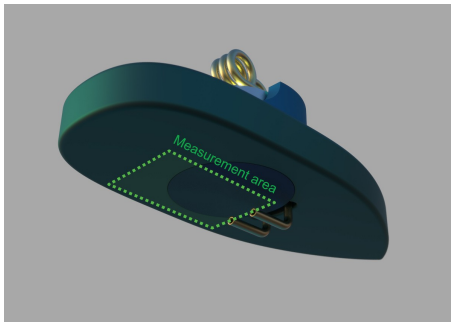
A spatially uniform but time-dependent velocity field is

$$u = U \cos(\omega t), \quad v = U \sin(\omega t),$$

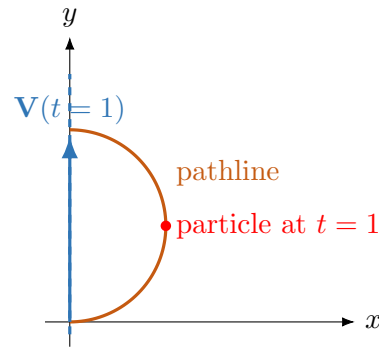
where $U = 2.0$ m/s and $\omega = \pi/2$ rad/s. A particle starts at the origin at $t = 0$. Determine:

1. the parametric pathline $x(t)$, $y(t)$;
2. the particle position at $t = 1.0$ s;
3. the direction and equation of the instantaneous streamline through the origin at $t = 1.0$ s;
4. why the streamline and pathline are different.

Particle-tracking context



Rotating-velocity diagram



Uniform rotating velocity: $U = 2.0$ m/s, $\omega = \pi/2$ rad/s; particle starts at the origin.

Hint

Integrate

$$\frac{dx}{dt} = U \cos(\omega t), \quad \frac{dy}{dt} = U \sin(\omega t)$$

from 0 to t . At $t = 1$ s, evaluate u and v . Because the velocity is spatially uniform, the instantaneous streamlines are straight lines parallel to the instantaneous velocity vector.

End of Flow Descriptions and Flow Lines Homework